MIL-P-14895E (AR)
25 April 1983
SUPERSEDING
MIL-P-14895D (AR)
25 September 1981

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, M509A1
METAL PARTS ASSEMBLY

This Specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, and packaging for Projectile, 8 Inch, M509A1 Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-6867 - Magnetic Inspection Units
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualificiation and Certification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1320

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures for Sampling by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

10551896 - Projectile, 8 Inch, M509A1
9229038 - Pallet for Projectile, 8 Inch, HE, M509A1

INSPECTION EQUIPMENT DRAWINGS

9278669 - Eddy Current Standard, Inspection M509 Type Base Plug

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-10 - Brinell Hardness of Metallic Materials
ASTM Method E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method E103 - Rapid Indentation Hardness Testing of Metallic Materials
ASTM Method B117 - Salt Spray (Fog) Testing
ASTM Method B247 - Aluminum-Alloy Die and Hand Forgings

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

AMS 2375 - Specification Control of Forgings Requiring First Article Approval

(Applications for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15906).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications.

3.2 Projectile body.

3.2.1 Forming. When a cold working operation is used in forming the projectile body, the maximum reduction in cross sectional area shall not exceed 22 percent. Process drawings showing the prior to cold work and after cold work process piece dimensions shall be submitted to cognizant engineering agency, for approval prior to use. Once approved, the process drawings submitted, shall not be changed without prior approval of the procuring activity. The hardness of the process piece prior to a cold working operation shall not exceed 241 Brinell. (see 6.7).

3.2.2 Stress relief. Prior to the band welding operation, projectile bodies which have been cold worked shall be stress relieved at 1000°F minimum temperature (body) for a minimum time of 1/2 hour.

3.2.3 Heat treatment. Each projectile body (After application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (See note 6.12).

3.2.3.1 Slow bend precracked charpy test. The cracked strength of the body shall be obtained for information purposes only.

3.3 Rotating band.

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3.3.1 Rotating band application. The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process in this instance means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary wire feed system. The process for applying the rotating band, once established must be maintained, controlled and monitored by the contractor. Procedures for maintaining, controlling and monitoring the process must be submitted to the technical agency for approval prior to the start of production. Documented records of the process control must be made available to the Government upon request. (See 6.9).

3.3.2 Rotating band seat. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface.

3.3.2.1 Process control. Process controls that will assure this feature are depicted on Dwg. 10551897. The contractor may, at his option, use controls other than the one defined on the drawing. If this option is excercised, the manufacturing procedure and the inspection plan for controlling the process shall be forwarded to the responsible technical agency for approval prior to its implementation into production.

3.3.3 Foreign matter on band seat. Prior to banding, the band seat should be free from any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.4 Metal defects.

3.4.1 Ogive, body and base. The ogive, body and base shall be free from all metal defects, in accordance with the defect standards (see 3.4.3) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits. (See 3.4.3)

3.4.2 Rotating band. The rotating band and band seat areas shall have no cracks, lack of bonding, or other metal defects in accordance with defect standards. (See 3.4.3).

3.4.3 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first months production or whenever requested by the procuring contracting agency.

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3.5 Proving ground function test. The assembled projectile metal parts shall reveal no evidence of failure when ballistically tested. Metal parts failure shall be:

a. Loss or breakup of metal part components inbore or in flight (except obturator).

b. Shear of the rotating band as evidenced by complete lack of engraving.

c. Base plug and cargo not ejecting when fired at service pressure and Zone 1. (Failure analysis must determine the cause to be the projectile metal parts).

3.6 Ogive and base.

3.6.1 Microstructure. The microstructure of the Ogive and Base shall comply with the MICROSTRUCTURE depicted in Figure 1a or 1b.

3.6.2 Heat treatment. Each base and ogive shall be solution heat treated and aged uniformly to meet the mechanical properties specified on the applicable drawing.

3.6.3 Monitoring of hot forge temperature. When bases and ogives are hot forged, the forger shall provide continuous, automatic monitoring of slug temperature prior to hot forging. Details shall be provided in accordance with paragraph 3.7.1.

3.6.4 Storage. Forgings, semi finished, or finished components shall be stored indoors. Forgings shall be processed into finished components in a sequential forging lot basis; i.e., first in first out.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.7.1 First article approval of aluminum forgings. First article approval of forgings, forging drawings, process and controls to be used in producing subsequent production forgings shall be obtained from the government technical agency prior to initiating production. The procedures for obtaining first article approval shall be in accordance with AMS 2375.

3.8 Control dimensions.

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3.8.1 Tool control dimensions. Dimensions marked tool control shall be gaged during First Article Testing and after eight hours maximum production on each machine in operation and prior to and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor.

3.8.2 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a .40% AQL.

3.9 Records of results of contractor conducted examinations and tests. The contractor shall maintain records of the results of all examinations and tests required by this specification and submit same upon request by the Government. (See 6.4).

3.10 Burrs and sharp edges. No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.11 Threads. Threads shall be full and undamaged, for the entire minimum length or depth as required by the applicable drawings.

3.12 Painting. Painting shall comply with the requirements of the applicable drawing.

3.13 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.13.1 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, or other foreign matter which may interfere with final inspection and completely free of same prior to shipping.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Projectile Metal Parts Assembly Complete with Bases	10551896	25
Projectile Subassembly	10551901	5
Body	10551897	5
Body (prior to Final Machining of Boat-tail)	10551897	5
Body (prior to banding)	10557897	2
Ogive	10551898	5
Ogive Forging (Prior to heat treat)	-	1
Base	10551899	5
Base Forging (Prior to heat treat)	-	1
Obturator	9287983	5

Twenty of the above projectile metal parts assemblies and bases shall be forwarded to a designated load plant for loading and shall then be forwarded to a government proving ground for ballistic testing.

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection. See MIL-A-48078.

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Table 1. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-14895E (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Projectile Metal Parts for Projectile, 8 Inch, M509A1		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ogive				
	Dwg. 10551898				
	Examination for defects	5	-	3.1	4.4.2.1
	Protective finish test	1	-	3.1	4.5.8
	Yield strength & elongation	2 specimens	-	3.6.2	4.5.1.5
	Microstructure	1	-	3.6.1	4.5.6
	Forging	*	-	3.7.1	*
	Obturator				
	Dwg. 9287983				
	Examination for defects	5	-	3.1	4.4.2.2
	Breaking load test	2	-	3.1	4.5.12
	Base				
	Dwg. 10551899				
	Examination for defects	5	-	3.1	4.4.2.3
	Protective finish test	1	-	3.1	4.5.8
	Yield strength & percent elongation	2 specimens	-	3.1	4.5.1.5
	Microstructure	1	-	3.6.1	4.5.6
	Forging	*	-	3.7.1	*

NOTES:

*See para 3.7.1 herein.

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TABLE 1. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-14895E (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body				
	Dwg. 10551897				
	Examination for defects				
	Prior to banding	2		3.1	4.4.2.4
	Prior to final machining of boat-tail	5		3.1	4.4.2.5
	After final machining	5		3.1	4.4.2.6
	Yield strength and percent elongation	2		3.2.3	4.5.1.4
	Band chemical analysis	2		3.1	4.5.4
	Projectile sub-assembly				
	Dwg. 10551901				
	Examination for defects	5		3.1	4.4.2.7
	Projectile metal parts assy				
	Examination for defects	25		3.1	4.4.2.8
	Protective finish test	2		3.12	4.5.8
	Ballistic test				
	a. Excess pressure	10		3.5	4.5.10.1
	b. Zone 1	5		3.5	4.5.10.2
	c. Service	5		3.5	4.5.10.3

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Ogives from not more than one forging manufacturer.
b. Bases from not more than one forging manufacturer.
c. Bodies from not more than one forging manfuacturer.

4.4.1.1 Body heat treatment lot. A heat treatment lot of bodies is defined as follows:

a. Standard procedure. A heat treatment lot of bodies shall consist of bodies from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process. The maximum lot size shall be 2000 bodies. If a mill heat exceeds 2000 bodies, the mill heat shall be divided into any convienent number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots.

b. Miscellaneous lot. Bodies manufactured from small heats of steel, or bodies that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1.a. The maximum lot size shall be 2000 bodies.

c. Use. The heat treatment lot constituted per 4.4.1.1.a and 4.4.1.1.b shall be the basis for selecting samples for mechanical properties tests. Bodies in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Heat treatment lot (ogive or base). A heat treatment lot of ogives or bases shall consist of components manufactured from material of the same alloy and temper from one manufacturer. The material shall be heat treated by the same process, in the same equipment and using the same temperature and time cycles. At no time will the lot quantity exceed 4800.

The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Components in each heat treatment lot shall remain segregated and identified until satisfactory completion of mechanical properties tests.

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4.4.1.3 - Miscellaneous lot (ogive or base). Components manufactured from small lots of material or those that have become separated from previously heat treated lots may be grouped into a miscellaneous lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.2. The lot thus constituted shall be the basis for selecting samples for the mechanical properties tests.

4.4.1.4 Projectile lot. Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging. Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14895E (AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER 10551898
4.4.2.1	Ogive		NEXT HIGHER ASSEMBLY 10551901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Metal defective		100%	3.4	Visual
b.	Hydrostatic	*		3.1	4.5.5
Major					
101	Runout of rear inside diameter with pitch diameter of body thread and rear shoulder		.40%	3.1	Gage
102	Runout of inside taper with pitch diameter of rear thread and rear shoulder - (2 places)		.40%	3.1	Gage
103	True position of diameter rear of fuze thread undercut with fuze thread		.40%	3.1	Gage
104	Pitch diameter of body thread, min.		.40%	3.1	Gage
105	Major diameter of body thread, min.		.40%	3.1	Gage
106	Length from rear face to forward inside basic diameter		.40%	3.1	Gage
107	Length from rear face to middle inside basic diameter		.40%	3.1	Gage
108	Length from rear face to rear inside basic diameter		.40%	3.1	Gage

NOTES:

* Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER 10551898
4.4.2.1	CONTINUATION SHEET		NEXT HIGHER ASSEMBLY 10551901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Diameter rear of fuze thread undercut		.40%	3.1	Gage
110	Pitch diameter of fuze thread		.40%	3.1	Gage
111	Minor diameter of fuze thread		.40%	3.1	Gage
112	Length from forward face to rear shoulder of fuze thread undercut		.40%	3.1	Gage
113	Length from forward face to forward outside basic diameter		.40%	3.1	Gage
114	Length from forward face to rear outside basic diameter		.40%	3.1	Gage
115	Width of fuze thread undercut, max.		.40%	3.1	Gage
116	Runout of projected fuze thread pitch diameter (2 places) with body thread pitch diameter, shoulder and forward face	**	.40%	3.1	Gage
117	Perpendicularity of shoulder to the body thread pitch diameter	**	.40%	3.1	Gage
118	Runout of outside diameter to body thread pitch diameter and shoulder at indicated length (2 places) from forward end	**	.40%	3.1	Gage
119	Yield strength and percent elongation	*	-	3.6.2	4.5.1.5
120	Microstructure	*	-	3.6.1	4.5.6
121	Protective finish	*	-	3.1	4.5.8

NOTES:

* Sample size as specified in applicable test method paragraph.

** Inspection advisory if Projectile Sub-assemblies are being procurred.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14895E (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 10551898
4.4.2.1	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 10551901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Inside diameter rear end		.65%	3.1	Gage
202	Length from front end to shoulder		.65%	3.1	Gage
203	Length total		.65%	3.1	Gage
204	Diameter of body thread undercut		.65%	3.1	Gage
205	Width of body thread undercut		.65%	3.1	Gage
206	Diameter of fuze thread undercut		.65%	3.1	Gage
207	Radii or chamfers missing or incorrect		.65%	3.1	Visual
208	Surface finish improper		.65%	3.1	Visual
209	Burrs and sharp edges		.65%	3.10	Visual
210	Threads damaged or not full		.65%	3.11	Visual
211	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14895E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287983
4.4.2.2	Obturator				NEXT HIGHER ASSEMBLY 10551896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		.40%	3.1	4.4.3.4/Gage
102	Thickness of band		.40%	3.1	4.4.3.4/Gage
103	Thickness at small outside diameter		.40%	3.1	4.4.3.4/Gage
104	Distance between two inside diameters		.40%	3.1	4.4.3.4/Gage
105	Length of small inside diameter		.40%	3.1	4.4.3.4/Gage
106	Breaking load test	*		3.1	4.5.12
107	Overall length		.65%	3.1	4.4.3.4/Gage
Minor					
201	Evidence of poor workmanship		.65%	3.13	4.4.3.4/Visual

NOTES:

*Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Base		10551899
			NEXT HIGHER ASSEMBLY 10551896

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.4.1	4.5.9
Special					
a	Diameter of press fit, min (3 equally spaced locations, forward and aft along length)		100%	3.1	Gage
Major					
101	Runout of outside diameter taper with press fit diameter (2 places)		.40%	3.1	Gage
102	Perpendicularity of shoulder with press fit diameter		.40%	3.1	Gage
103	Length from shoulder to rear outside basic diameter		.40%	3.1	Gage
104	Length from shoulder to forward outside basic diameter		.40%	3.1	Gage
105	Length from shoulder to front face		.40%	3.1	Gage
106	Thickness from bottom of cavity to front face		.40%	3.1	Gage
107	Diameter press fit, max		.40%	3.1	Gage
108	Radius between press fit diameter and shoulder		.40%	3.1	Gage
109	Yield strength and percent elongation	*		3.6.2	4.5.1.5
110	Microstructure	*		3.6.1	4.5.6
111	Protective finish	*		3.1	4.5.8

NOTES:

* Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET	2 OF 2	10551899
			NEXT HIGHER ASSEMBLY 10551896

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth from cavity basic diameter to bottom of cavity		.65%	3.1	Gage
202	Length from rear surface to shoulder		.65%	3.1	Gage
203	Cavity radius		.65%	3.1	Gage
204	Radii or chamfers missing		.65%	3.1	Visual
205	Surface finish improper		.65%	3.1	Visual
206	Burr		.65%	3.10	Visual
207	Painting incomplete or damaged		.65%	3.12	Visual
208	Paint on indicated surface		.65%	3.1	Visual
209	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Body (Prior to banding)		10551897
			NEXT HIGHER ASSEMBLY 10551896

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective in band seat area		100%	3.4.2	Visual
Major					
101	Runout of rear outside diameter with band seats		.40%	3.1	Gage
102	Diameter of band seats (2)		.40%	3.1	Gage
103	Length from front end of forward band seat flat to rear end of rear band flat		.40%	3.1	Gage
Minor					
201	Radii or chamfers missing		.65%	3.10	Visual
202	Burr or sharp edges		.65%	3.1	Visual
203	Surface finish or band seat improper		.65%	3.3.3	Visual
204	Evidence of poor workmanship		.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 5	DRAWING NUMBER
4.4.2.5	Body (Prior to final machining of boat-tail)		10551897
			NEXT HIGHER ASSEMBLY 10551901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Band seat metal defective	*	3.4.2	4.5.3.1	
b.	Diameter of counterbore, max (3 equally spaced locations, forward and aft along length		100%	3.1	Gage
Major					
101	Length from front end of forward rotating band to rear end of rear rotating band		.40%	3.1	Gage
102	Width of forward rotating band		.40%	3.1	Gage
103	Width of rear rotating band		.40%	3.1	Gage
104	Runout of rear counterbore diameter with inside diameter at indicated depth from rear shoulder		.40%	3.1	Gage
105	Perpendicularity of rear face with counterbore diameter		.40%	3.1	Gage
106	Length from rear face to front end of forward rotating band		.40%	3.1	Gage
107	Pitch diamter of ogive thread, max		.40%	3.1	Gage
108	Minor diameter of ogive thread, max.		.40%	3.1	Gage

NOTES:

* Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER 10551897
4.4.2.5	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 10551901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Diameter of counterbore, forward end, min		.40%	3.1	Gage
110	Depth of counterbore, forward end		.40%	3.1	Gage
111	Length of dovetail groove, small		.40%	3.1	Gage
112	Length of dovetail groove, large		.40%	3.1	Gage
113	Length total		.40%	3.1	Gage
114	Length from front face to outside basic diameter at rear of ogive		.40%	3.1	Gage
115	Length from front face to outside basic diameter at forward end of ogive		.40%	3.1	Gage
116	Length from rear end to inside basic diameter at rear of ogive thread		.40%	3.1	Gage
117	Relative location of dovetail grooves with each other		.40%	3.1	Gage
118	Uninterrupted surface on inside diameter (longitudinal)		.40%	3.1	Visual/Manual
119	Body cavity surface finish improper		.40%	3.1	Gage
120	Runout of ogive with bourrelets (2 places)		.40%	3.1	Gage
121	Runout of outside diameter between bourrelets with bourrelets		.40%	3.1	Gage
122	Runout of inside diameter with bourrelets		.40%	3.1	Gage
123	Runout of outside diameter between rotating bands with bourrelets		.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 5	DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET		10551897
			NEXT HIGHER ASSEMBLY
			10551901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
124	Runout of forward rotating band diameter, at basic length, with bourrelets		.40%	3.1	Gage
125	Runout of rear rotating band diameter with bourrelets		.40%	3.1	Gage
126	Diameter of counterbore, rear end		.40%	3.1	Gage
127	Diameter forward of boat-tail undercut		.40%	3.1	Gage
128	Diameter outside between rotating bands		.40%	3.1	Gage
129	Diameter of bourrelets, min.		.40%	3.1	Gage
130	Diameter, at basic length, of forward rotating band		.40%	3.1	Gage
131	Diameter of rear rotating band		.40%	3.1	Gage
132	Depth from basic chord to bottom of dove tail groove (small)		.40%	3.1	Gage
133	Depth from basic chord to bottom of dove tail groove (large)		.40%	3.1	Gage
134	Width of dove tail groove, at intersection with inside diameter, max (small)		.40%	3.1	Gage
135	Width of dove tail groove, at intersection with inside diameter, max (large)		.40%	3.1	Gage
136	Width of dove tail groove at basic length (large)		.40%	3.1	Gage

NOTES:

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PARAGRAPH	TITLE	SHEET 4 OF 5	DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET		10551897
			NEXT HIGHER ASSEMBLY 10551901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
137	Width of dove tail groove at basic length (small)		.40%	3.1	Gage
138	Angle of forward rotating band		.40%	3.1	Gage
139	Length of counterbore, rear end of body		.40%	3.1	Gage
140	Runout forward of boat-tail undercut with bourrelets		.40%	3.1	Gage
141	Wall thickness of forward end of dovetail groove (2) min.		.40%	3.1	Gage
142	Hardness test of body		100%	3.2.3	4.5.1.2
143	Yield strength and percent elongation	*		3.2.3	4.5.1.4
144	Band chemical analysis	*		3.1	4.5.4
145	Inside diameter of body cavity		.40%	3.1	Gage
146	Runout of the projected ogive thread pitch diameter with bourrelets and front face at the two indicated lengths	**	.40%	3.1	Gage
147	Perpendicularity of ogive thread pitch diameter with forward face	**	.40%	3.1	Gage

NOTES:
*Sample size as specified in applicable test method paragraph.
**Inspection advisory if Projectile Sub-assemblies are being procurred.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET				10551897
					NEXT HIGHER ASSEMBLY
					10551901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Outside diameter between bourrelets		.65%	3.1	Gage
202	Width of obturator groove		.65%	3.1	Gage
203	Length from rear end of body to forward end of rear bourrelet		.65%	3.1	Gage
204	Length from front face to rear of forward bourrelet		.65%	3.1	Gage
205	Diameter of obturator groove		.65%	3.1	Gage
206	Width of rotating band undercuts (3), if applicable		.65%	3.1	Visual/Gage
207	Depth of rotating band undercuts (3), if applicable		.65%	3.1	Visual/Gage
208	Burrs or sharp edges		.65%	3.10	Visual
209	Radii or chamfers missing		.65%	3.1	Visual
210	Threads damaged or not full		.65%	3.11	Visual
211	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10551897
4.4.2.6	Body				NEXT HIGHER ASSEMBLY 10551901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.4.1	4.5.3.2
Special					
a.	Max. radius between rearface and counterbore diameter		100%	3.1	Gage
b.	Location of index mark with keyway		100%	3.1	Gage
c.	Burrs in counterbore		100%	3.1	Visual
Major					
101	Runout of boat-tail with bourrelets at indicated length from rear face		.40%	3.1	Gage
102	Length from rear end to outside basic diameter on boat-tail		.40%	3.1	Gage
Minor					
201	Location of marking		.65%	3.1	Gage
202	Marking missing, incorrect, or illegible		.65%	3.1	Visual
203	Burrs or sharp edges		.65%	3.10	Visual
204	Protective coating inadequate or damaged		.65%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14895E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10551901
4.4.2.7	Projectile Sub-Assembly				NEXT HIGHER ASSEMBLY 10551896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Magnetic particle stamp missing (indicating failure to pass test)		100%	3.1	Visual
Major					
101	Runout of projected pitch diameter of fuze thread with bourrelets and front face (2 places)		.40%	3.1	Gage
102	Runout of forward end of ogive with bourrelets		.40%	3.1	Gage
103	Runout of rear end of ogive with bourrelets		.40%	3.1	Gage
104	Length total		.40%	3.1	Gage
105	Gap between ogive and body		.40%	3.1	Gage
106	Air leak test		100%	3.1	4.5.7
Minor					
201	Burrs or sharp edges		.65%	3.10	Visual
202	Threads damaged or not full		.65%	3.11	Visual
203	Protective coating inadequate or damaged		.65%	3.1	Visual
204	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Projectile 8 Inch HE, M509A1 Metal Parts Assembly	1 OF 1	10551896
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Ballistic test				
	a. Excess pressure	*		3.5	4.5.10.1
	b. Zone 1	*		3.5	4.5.10.2
	c. Service pressure	*		3.5	4.5.10.3
Major					
101	Airleak stamp missing (indicating failure to pass test)		100%	3.1	Visual
102	Protective finish	*		3.12	4.5.8
103	Diameter of bourrelets, max.		100%	3.1	Gage
104	Rotating band damaged		.40%	3.1	Visual
105	Obturator missing or damaged		.40%	3.1	Visual
106	Threads damaged		.40%	3.11	Visual
107	Protective finish incomplete or damaged		.40%	3.12	Visual
108	Interior surface coating missing or incomplete		.40%	3.1	Visual

NOTES:

*Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10551896
4.4.2.8	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Protective caps (fuze thread & base) missing		.65%	3.1	Visual
202	Paint on rotating band bearing surface		.65%	3.12	Visual
203	Paint on obturator		.65%	3.12	Visual
204	Paint not dry		.65%	3.12	Visual/Manual
205	Paint on threads		.65%	3.12	Visual
206	Marking missing, incorrect or illegible		.65%	3.1	Visual
207	Burrs or sharp edges		.65%	3.10	Visual
208	Grommet missing or loose		.65%	3.1	Visual/Manual
209	Evidence of poor workmanship		.65%	3.13	Visual

NOTES:

* Not required for intraplant use.

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Tables, and herein.

4.4.3.1 Hardness prior to cold forming (major defect). When the contractor elects to cold form the body he will assure compliance with paragraph 3.2.1. Bodies subjected to this test will be hardness tested at a midpoint along the length of outside diameter of the body. For a minimum of 20,000 bodies, each body will be tested. Based upon conformance with the requirement during this testing the contractor may propose an alternate hardness sampling procedure. As a minimum the alternate hardness sampling procedure shall include hardness testing of the first twenty consecutive bodies out of slow cool after each start up and sampling for hardness on a specific hourly basis. The alternate procedure shall be submitted to the procuring activity together with supporting hardness data for approval. (See 4.5.1.1).

4.4.3.2 Verification of base heat treat (critical defect). After heat treat each base shall be verified for its having been heat treated. Bases failing to comply will be rejected. Rejected parts may be heat treated and resubmitted. All bases passing this test shall be stamped in accordance with applicable drawing. Testing shall be in accordance with para. 4.5.1.3.

4.4.3.3 Slow bend precrack charpy (information only). Projectile bodies selected for testing in accordance with 4.5.1.4 for high hardness shall have four (4) specimens taken and tested per 4.5.11. The four specimens shall be selected in the longitudinal direction with centers located approximately 4 inches from the rear of the body. Two (2) samples each shall be tested at minus sixty-five degrees fahrenheit ($-65^{o}F$) and the other two (2) at seventy degrees fahrenheit ($+70^{o}F$). The notches shall be machined on the outside surface of the specimen.

4.4.3.4 Dimensional control of molded parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1), a sample of six (6) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts shall be performed on the above quantities from each cavity after production of each 1000 parts or at the end of each shift, providing the number of parts produced during the shift is less than 1000 parts. Of the six (6) samples, five (5) samples shall be the last parts produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

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If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be rejected.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

The contractor shall furnish evidence to the Government inspector that dimensional stability has been attained prior to measuring the parts for acceptance.

NOTE 1: Establishing a curing time as follows. Dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to: Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification and/or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of item produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.4.2 Certification of nondestructive test personnel. All personnel operating or calibrating nondestructive test equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

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4.5.1.1 Hardness prior to cold forming. Prior to testing, the surface of each body, in the area to be tested, shall be prepared by machining or grinding a flat of sufficient depth, .030 inches minimum, to remove scale and get below the decarburation zone. Test shall be performed using the procedure of ASTM Method E-103. Bodies failing to comply with 3.2.1 may be reannealed and retested.

4.5.1.2 Hardness (body). In order to determine uniformity of heat treatment each body after application of the welded overlay band and final heat treatment and before final machining shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the body approximately one inch forward of the leading rotating band. The test shall be made in accordance ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. (Note: Flats and impressions shall be completely removed by final machining.)

b. Hardness tester periodic verification shall be performed using brinnel test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

d. The hardness tester shall use a standard 10mm brinnel ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the body to be tested with a 10,000-500 lb clamping load. Second, apply a minor load of approximately 10% of the 3000 kg of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetrator travel from initial minor load position to new minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the body.

4.5.1.3 Verification of heat treat of base. Testing shall be with eddy current or any other approved method.

4.5.1.4 Yield strength and percent elongation of body. The two (2) hardest and two (2) softest bodies from each standard heat treatment lot or the four (4) hardest and four (4) softest bodies from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.2 shall be subjected to this test. Samples

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for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. Two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the band seat area, and two specimens as prescribed at ASTM Method E-8 shall be taken longitudinally from opposite sides of the body, located approximately seventeen (17) inches from the rear end to the center of the specimen. The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements.

The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor the entire lot may be reheat treated and retested; or a new set of bodies may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be reheat treated and retested.

4.5.1.5 Yield strength and percent elongation of ogive and base. The sample size for the tension test shall be one component (each base and ogive) from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The base specimens shall be taken from midwall centered between rear and front ends 180° apart. The ogive tension test specimen shall be taken from the midwall at a location where the gage length center of the specimen is approximately 1 inch from the finished rear face of the body thread end of the ogive, 180° apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample.

4.5.2 Rotating band seat locations. The contractor may use any control method that will assure the band seats meet the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.4.2.4. When the contractor establishes a control method, the

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manufacturing procedure, and inspection plan shall be forwarded to the Technical Agency for approval prior to use. No change to these controls will be made once established without approval from the Technical Agency.

4.5.3 Inspection for body metal defects after machining.

4.5.3.1 Magnetic particle inspection of band seat. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable. (See 6.9).

The samples provided shall have the band removed by machining the band and chemically etching without damaging the base metal (steel) of the band seat (see 6.8). Each body sampled shall then be visually inspected for compliance with para 3.4.2 using the magnetic particle inspection technique.

The procedure for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method with an indirect magnetization by either the continuous or residual method, inducing a circular magnetic field, and again after inducing a longitudinal magnetic field.

Should two consecutive failures occur in any welding machine(s), in any reduced sample size, the machine(s) shall revert back to the original first five consecutive banded bodies per that welding machine(s) and after five consecutive acceptable bodies, the sequence of reduced sampling will also apply. Banded bodies shall be identified with the welding machine that applied the band.

4.5.3.2 Finished body. Projectile bodies after final machining shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-I-6868, for wet fluorescent method with indirect magnetization by the continuous method. The entire body, inside and outside, shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. A magnetic particle standard shall be utilized to establish inspection sensitivity. The design for the standard shall be submitted for approval prior to use (see Para. 4.4.4). Any body which fails to comply with the requirements of 3.4.1 shall be rejected. Bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

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4.5.3.3 Magnetic particle suspension. The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted per 4.5.3.2 since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867.

4.5.4 Band chemical analysis. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been tested and found acceptable. Reduced sampling of one in each 4 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been tested and found acceptable. Sample chips shall be obtained from each band from the last .030 inches removed when machining to the finished diameter. All chips shall be collected and identified with the band and shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet.

Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

4.5.5 Hydrostatic test of ogive. Prior to the application of protective finish and gaging, the finish machined ogive shall be subjected to a hydrostatic test. The ogive shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the test pressure was actually lower than that specified on the applicable drawing, the ogives tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. The fixture should be such that it does not axially restrain the ogive.

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4.5.5.1 Lot sampling plan. One hundred percent (100%) hydrostatic testing of finish machined ogives shall be conducted until 1200 ogives consecutively pass the hydrostatic test. ubsequently lots of no greater than 5000 ogives may be submitted for testing. A sample of 50 ogives per lot shall then be tested. Failure of any ogive to meet the requirement of the applicable drawing shall be cause for rejection of the lot represented by the sample. When a lot has been rejected while on sampling inspection, the frequency shall revert back to the 100 percent inspection until 1200 ogives consecutively pass the hydrostatic test, to qualify for reduced sampling testing.

4.5.5.2 Continuous sampling (option). The ogive hydrostatic testing may be conducted using a continuous sampling plan from MIL-STD-1235, CSP-1 with AQL of 0.10.

4.5.6 Microstructure examination (ogive & base). One ogive & one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's Reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous or continuous network form shall be cause for rejection of the lot represented by the sample.

Two examples of acceptable microstructures are shown in Figures 1a & 1b. In Figure 1a, the insoluble phase is well broken-up randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1b, the insoluble phase is well broken-up and randomly distributed within a matrix which is partially recrystallized. Two examples of the unacceptable microstructures are shown in Figure 2a and 2b. In Figure 2a, the insoluble phase is distributed in a semi-continuous form within the matrix which shows microsegregation (or coring). In Figure 2b, the insoluble phase is distributed in a more continuous form (network) within a matrix which shows microsegregation (or coring). Either of the above structures (Figure 2a & 2b) must be avoided because they are detrimental to the integrity of the material.

Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys.

4.5.6.1 Ogive specimen: A metallographic specimen at least 1/2 inch in length shall be removed from the sidewall. It shall be a longitudinal cross section of the body thread area with the center of the specimen approximately 3/4 inches from the finished rear face.

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4.5.6.2 Base specimen: A metallographic specimen at least 1/2 inch in length shall be removed from the base and be a longitudinal cross section with its center located at mid-diameter of the base.

4.5.7 Air leak test. The ogive to body joint of the projectile subassembly shall be tested in accordance with the applicable drawing for 15 seconds minimum. A force shall not be applied in the axial direction such as to compress the ogive to body joint. Failure to comply shall be cause to reject the projectile subassembly. Projectile subassemblies which comply shall be stamped with the letter A in accordance with the applicable drawing. The test may be accomplished by one of the following procedures.

4.5.7.1 Submersion test (alternate 1). Metal parts subassemblies shall be mounted in an approved fixture in such a manner that the subassembly shall be submerged in water. The required air pressure shall be applied and held for the minimum specified time, after the water in the tank is permitted to settle after immersion. Observation for leakage shall be made for the entire time the metal parts subassembly is required to be pressurized. The presence of air bubbles in the vicinity of the body to ogive joint or rising to the surface of the water shall be cause for rejection of the metal parts subassembly.

4.5.7.2 Pressure drop test (alternate 2). Each projectile sub-assembly shall be mounted so as to seal the projectile above and below the ogive to body joint with a collar (5 in^3 max.) and pressurizing this collar at 15 psi for 15 seconds while looking for a maximum pressure drop of 0.01 psi during the 15 second hold cycle. Pressure source shall be inactive during the hold cycle. A pressure drop in excess of 0.01 psi shall be cause for rejection of the subassembly.

4.5.8 Protective finish test. A salt spray test shall be performed on each of the following; projectile metal parts assembly, ogive and base utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable detailed specification for the protective finish, except the curing cycle for the assembly shall be in accordance with component dwg. 10551896. Testing shall be performed on one ogive and base for each twenty-four hours of continuous production or fraction thereof. The sample size for the assembly shall be as specified in the applicable specification. Failure to comply will be cause for rejection of the production since the previous test.

4.5.9 Eddy current inspection of base. The base, after machining and prior to the application of any protective coating, shall be inspected for metal defects using the eddy current method. Bases failing to comply with requirements of 3.4.1 shall be rejected. Standards and test method shall be in accordance with Drawing 9278669. Each acceptable base shall be stamped with the letters "EC" on the rear most surface of the base.

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4.5.10 Proving ground ballistic tests. The following tests shall be conducted using cannon tubes having 25 percent minimum remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exits the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.5. Failure shall be cause for rejection of the lot represented by the sample. Range, deflection, and velocity shall be recorded for each round fired for information. An attempt will be made to recover the projectiles fired for this test.

4.5.10.1 Excess pressure. The metal parts assemblies supplied for this test shall contain inert cargo with the base assembled and fuzed with a dummy fuze and dummy explusion charge assembly, will be fired for recovery in an M201A1 Cannon. A Zone 9 M188A1 Propellant Charge conditioned to +145°F with propellant added as necessary to produce a pressure of 44,300 to 48,100 PSI shall be used.

4.5.10.2 Zone 1 (-40 plus or minus 5 F). The metal parts assemblies supplied for this test shall contain inert cargo with the base assembled and fuzed with applicable fuze and expulsion charge assembly and conditioned to -40 ± 5°F will be fired in an M201A1 cannon. A zone 1 propellant charge producing an approximate pressure of 9,000 to 10,000 PSI shall be used. The metal parts assemblies will be fired for accuracy at a range selected by the proof director.

4.5.10.3 Service pressure. The metal parts assemblies supplied for this test shall contain inert cargo with the base assembled and fuzed with applicable fuze and expulsion charge assembly will be fired in an M201A1 cannon. A zone 9 propellant charge producing an approximate pressure of 36,500 to 38,500 PSI (service pressure) shall be used.

4.5.10.4 Sampling procedure. Normal sampling will be conducted until 5 consecutive lots have been accepted. Reduced sampling shall then be instituted. Any failure, while on reduced sampling, mandates return to normal sampling. Sample sizes for normal and reduced sampling shall be as follows:

	Normal	Reduced
a. Excess pressure	5	2
b. Zone 1	3	3
c. Service pressure	2	-

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4.5.11 Slow bend precrack charpy test. Testing shall be conducted in accordance with the test procedures in the Appendix A of this specification. The contractor shall submit a copy of all test reports monthly, reporting all information required by the Appendix.

4.5.11.1 Sample. The four specimens shall be selected in the longitudinal direction with centers located approximately 4 inches from the rear of the body. Two (2) samples each shall be tested at minus sixty-five degrees fahrenheit (+70°F). The notches shall be machined on the outside surface of the specimen.

4.5.12 Obturator breaking load test. A sample of twenty (20) obturators shall be randomly selected from each lot of obturators. If any of the samples fail to meet the requirements of Dwg. 9287983, the lot shall be rejected. The obturator shall be tested for breaking load using an approved test fixture. The test fixture shall consist of two one-inch diameter opposing spindles machined to the Internal profile of the obturator. The obturator shall be assembled and tested using a test speed of 2.00 inches per minute.

5. PACKAGING.

5.1 Packing. This section (Section 5) is not applicable if LAP operations are to be performed at the same facility.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C - Minimum Military Pack.

5.2 Packaging and packing.

5.2.1 Projectile metal parts assemblies shall be palletized in accordance with Drawing 9229038.

5.2.2 Bases shall be packaged and packed in accordance with MIL-STD-1169, and will conform to contractor drawings.

5.3 Marking. Marking of the pallet shall comply with Dwg. 8837839. Marking of the cartons for the bases shall comply with MIL-STD-1169 and shall include marking to assure that the base is shipped or stored with the cavity facing downward. Bases need not be identified with projecitle lot numbers.

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6. NOTES.

6.1 Intended use. The projectile metal parts covered by this specification are intended for the Projectile, 8 Inch, M509A1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be develolped as specified by an approved Data Item

Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are involked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID
3.2.1, 3.2.3, 3.3.1, 3.3.2.1, 3.7.1, 3.8.2, 4.4.3.1, 4.5.2, 6.12	Procedure Report	* DI-T-5372 (Tailored)
4.5.1.1, 4.5.1.2, 4.5.1.4, 4.5.1.5, 4.5.4, 4.5.5, 4.5.6, 4.5.7, 4.5.8, 4.5.12	Quality Inspection Test Demonstration & Evaluation Report	DI-R-1724
4.5.1.3, 4.5.3.1, 4.5.3.2, 4.5.9	Quality Inspection Report	DI-R-1721
4.5.11	Test and Demonstration Report	DI-T-1906

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Cold forming. The contractor is cautioned that if cold forming is utilized in processing of the projectile body, those bodies which do not completely clean up in the machining operation prior to the cold forming have a greater propensity to develop circumferential tears during the forming operation particularly if areas of non cleanup are heavily scaled.

6.8 Suggested etching procedures for removing gilding metal overlay.

a. Recommended. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.005 to 0.010 inch thick will dissolve in 8 to 12 hours. Renew solution as necessary.

b. Solution. Five hundred (500) grams Chromic oxide CrO_3 (commercial flakes) 50 ml. sulphuric acid H_2SO_4 (technical grade) 950 ml. water H_2O.

This solution may be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.005 to 0.010 inch thick will require approximately two to three hours to dissolve in a fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

6.9 Band welding. Recommend welding operation of overlay welded band be guided by Frankford Arsenal "Process Recommendations for Banding 155MM, M483 Projectile" dated 19 April 1973. Requests for copies should be submitted to: Commander, ARRADCOM, ATTN: DRDAR-LCU-M, Dover, NJ 07801.

6.10 Steel blooms, billets, slabs and bars.

6.10.1 Raw material. The supplier of the steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

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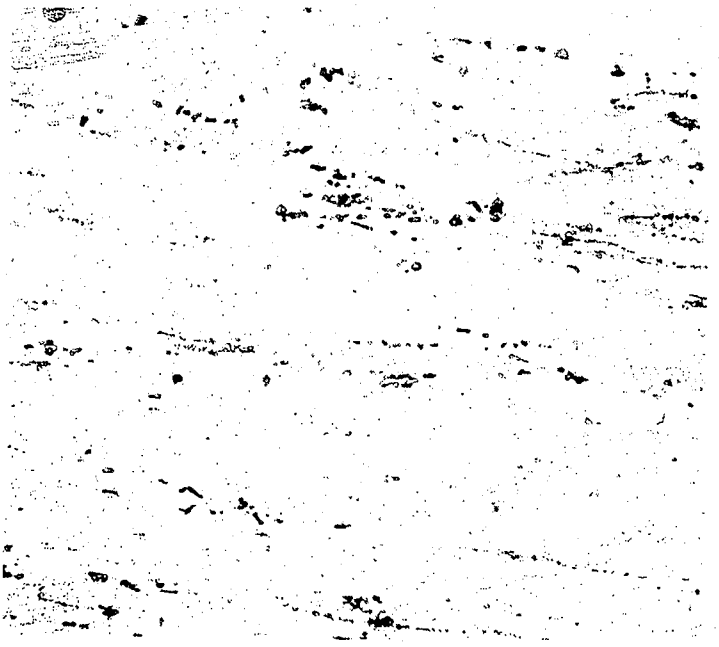
6.10.2 Forging quality processing techniques. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.11 Adequate working of ogive material. In order to insure adequate working of the ogive material and the break-up of insolubles from the cast ingot to the final raw forging configuration, it is recommended that a minimum extrusion ratio of six (6) be used representing an 83% reduction of area from the cross-sectional area of the extrusion.

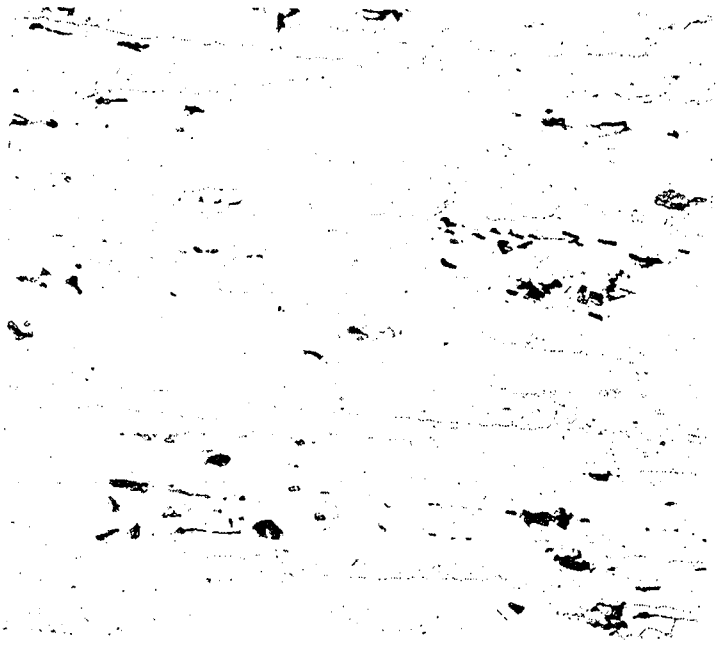
6.12 Heat treatment procedure. Contractor shall submit, for information purposes only, the heat treatment procedure per DI-T-5372.

6.13 Preproduction sample/hardness range. To minimize the need for reheat treating material or for having miscellaneous heat treatment lots, it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill heat. When using a hardness range, acceptance will still be in accordance with paragraph 4.5.1.4.

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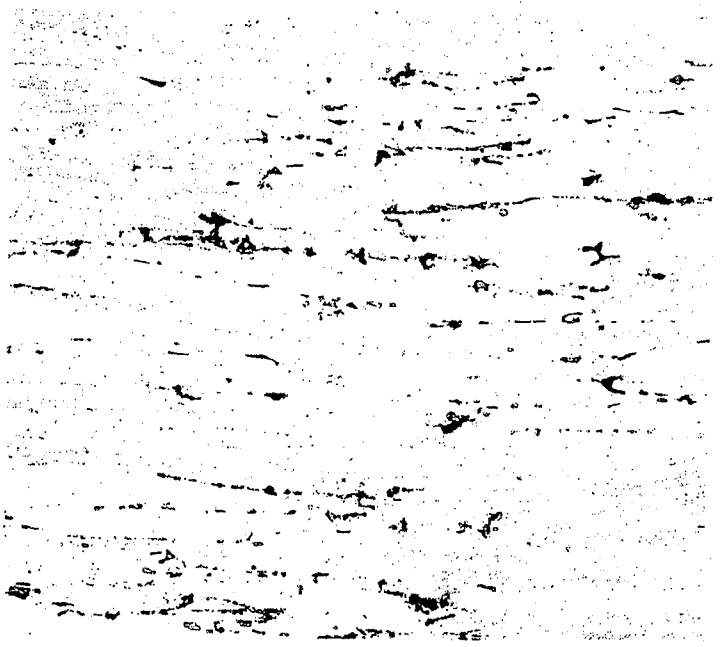
a



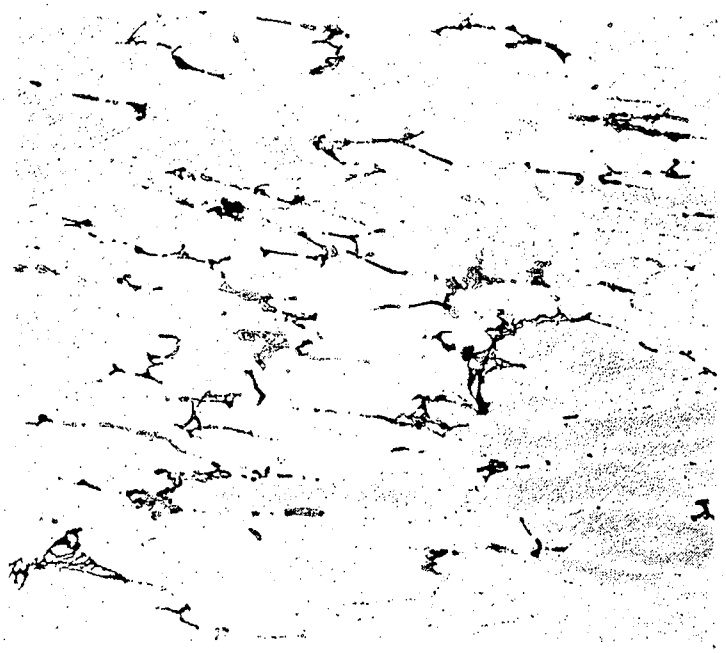
b

FIGURE 1 Acceptable Microstructures (250x)

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a



b

FIGURE 2 Unacceptable Microstructures (250x)

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Preparing activity:
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(Project 1320-A630)

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APPENDIX

TEST METHOD FOR SLOW BEND PRECRACKED CHARPY TEST

10. SCOPE

10.1 Scope. This appendix covers the test method for determining fracture toughness using a Charpy specimen containing a fatigue crack and tested in slow bending. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 ASTM STANDARDS

E 4 Verification of Testing Machines

E 139 Recommended Practice for Conducting Creep, Creep-Rupture and Stress Rupture Tests of Metallic Materials

E 399 Plane Strain Fracture Toughness of Metallic Materials

30. SUMMARY OF METHOD

30.1 The method involves three point bend testing of notched and fatigue cracked specimens using fixtures which minimize the contribution of friction forces to the measured applied load. The load-deformation characteristics of the test is recorded and the fracture toughness is determined from these characteristics and the original dimensions of the specimen.

40. PRECAUTIONARY NOTES

40.1 Precautionary notes. When preparing specimens and testing, it is important that certain precautions be kept in mind.

40.1.1 Crack strength. The crack strength of some materials will be sensitive to the rate of loading and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The crack strength will be influenced by the testing temperature and calibration relations may change with testing temperature.

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APPENDIX

50. DEFINITIONS

50.1 a - fatigue crack depth, Inches (In)

B - specimen thickness (In)

W - specimen depth (In)

M - Bending Moment, Inches-Pound Force (In - lb_f)

f(a/W) - Power series of ratio (a/W) from ASTM-E-399

S - Span between support rollrs; (In.)

P max - Maximum Load Applied - (lb)

P_p - A specific point at which the first load instability indicating unstable crack growth is recorded (lbs.)

K - Stress intensity factor (Ksi IN)

S - Span between support rollers - (In)

K_f (max) - Maximum stress intensity factor in fatiguing (KSI In)

K_Q - Calculated fracture toughness (KSI IN.)

K_{IC} - Plane strain fracture toughness (KSI IN).

60. APPARATUS

60.1 Tension testing machine. Tension Testing Machine conforming to the requirements of ASTM E 4.

60.2 Bend-test fixture. The procedure involves the testing of the specimen in three point bending. The bend fixture should be so designed as to minimize errors that arise from friction between the specimen and the supports. This can be accomplished by allowing the support rollers to rotate and move apart slightly during loading as described in ASTM E 399. The nominal span shall be 1.97 in. If the diameter of the support rollers is between 0.30 in. to 0.40 in. there will be a negligible change in the span during deformation of the specimen.

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60.3 Temperature control. When testing at -65°F, an integral temperature conditioning chamber shall be used. Any suitable method may be used to temperature condition the specimen providing the region including the notched and cracked section can be uniformly maintained within ± 2°F of the desired test temperature. The cooling medium shall be free of contaminants, such as chloride temperature. The cooling medium shall be free of contaminants, such as chloride or water. When testing a 70°F, no chamber is necessary, but the temperature must be held to plus or minus 8°F.

60.4 Temperature measurement. The device used to measure the temperature of the medium should be placed adjacent to the test specimens. If an agitated liquid medium is used, the specimens shall be held at the desired temperature, plus or minus 2°F, for at least 10 minutes. Specimens being tested in gas shall be held at the desired temperature, plus or minus 2°F, for at least 60 minutes. See ASTM-E-139 for a discussion of errors on measurements.

60.4.1 Calibration. The temperature measuring apparatus should be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS AND PREPARATION

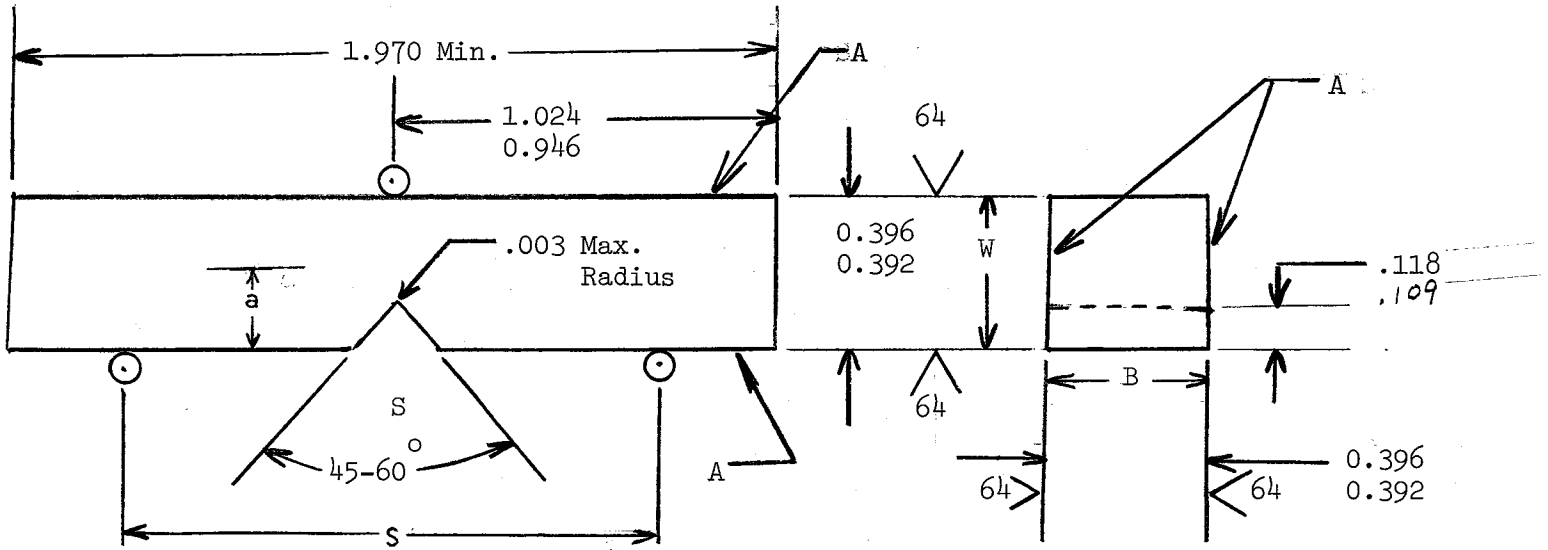
70.1 Dimensions. The precracked Charpy bend specimen is shown in Fig. 1. The crack depth (including notch), (a) is nominally equal to one-half the thickness, (b) and shall lie within the range between 0.45 to 0.55 B.

70.1.1 Crack starter. The crack starter may consist of a "V" notch having an included angle between 45° and 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. and minimum machining surface compression stress is necessary in order to facilitate the inititation of a fatigue crack at a low stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed.

70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

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NOTE 1: All dimensions in inches.

NOTE 2: "A" surfaces shall be perpendicular or parallel as applicable to within 0.001 in TIR.

NOTE 3: Crack starter shall be perpendicular to specimen length and thickness to within ± 2 degrees.

FIGURE 3 Dimensions of Precracked Charpy Specimens

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70.2.2 Cracking method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three point bending may be used to produce fatigue cracks. Cantilever bending facilitates completely reversed loading which aids in starting and propagating the crack. For practical purposes, the K calibration for three point bending may be used to estimate the stress intensity factors.

The expression given in ASTM E 399 for the K calibration in three point bending may be rewritten in terms of the bending moment M as follows and used to compute the stress intensity in cantilever bending: $K = (4M/BW^{3/2})\, f(a/W)$ where the values of the function $f(a/W)$ are given in tabular form in ASTM E 399.

The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane normal to the notched face of the specimen.

70.2.3 Cracking. During the last 0.020 in. of fatigue cracking, the maximum intensity of the fatigue cycle, K_f (max) shall not exceed $0.6K_Q$. For low temperature toughness test requirements the K_f shall be calculated using the low temperature K_Q. The stress intensity range should not be less than $0.9K_f$ (max).

80 TEST PROCEDURE

80.1 Number of tests. Samples shall be selected as specified in the technical data package.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Fig. 1.

80.2.1 Measure the thickness, (B) to the nearest 0.001 in. at not less than two positions, one on each side of the crack starter notch and record the average value.

80.2.2 After fracture measure the initial fatigue crack depth, (a) to the nearest 0.5 percent at the following three positions: at the center of the crack front and midway between the center and the inter-section of the crack front with the specimen surfaces. Use the average of these three measurements as the crack depth in the calculation of K_Q (see 90.1). If the difference between any two

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of the crack length measurements exceeds 5 percent of the average, or if part of the crack front is closer to the machined notch root than 5 percent of the average, the specimen should be discarded. Also, if the length of either surface trace of the crack is less than 80 percent of the average crack length, as defined above, the specimen should be discarded.

80.3. Bend testing. Set up the bend test fixture so that the line of action of the applied load shall pass midway between the support roll centers within 0.025 in. of the distance between these centers. Measure the span (distance between the support roll centers) within 1 percent and record this value as S. This value should fall between 1.49 in. and 1.65 in. If not, the span should be adjusted to fall within this range.

80.3.1. Locate the specimen with the crack tip midway between the roll centers to within 0.025 in., and square to the roll axis within 2 deg. Check the parallelism between the back face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller should be suitably adjusted.

80.3.2. Load the specimen at a rate between 700 lb_f/min and 3600 lb_f/min until the maximum load is reached and autographically record the specimen load-displacement characteristics.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 Calculations. Calculate the fracture toughness of the specimen in compliance with the method and requirements outlined in ASTM-E-399. If a pop-in occurred, the minimum pop-in load P_p will be used as the basis for the calculation. For the bend specimen, calculate K_Q in units of psi-in $^{1/2}$ as follows:

$$K_Q = (P_p\ S/BW^{3/2})\ 2.9\ (a/W)^{1/2} - 4.6$$

$$(a/W^{3/2} + 21.8\ (a/W)^{5/2} - 37.6\ (a.W)^{7/2} + 38.7\ (a/W)^{9/2})$$

$$K_Q = P_p\ S/BW^{3/2}\ f\ (a/W)$$

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APPENDIX

The power series f (a/W) are tabulated in the following table for specific values of a/W:

a/W	f(a/W)	a/W	f(a/W)
0.450	2.28	0.500	2.66
0.455	2.32	0.505	2.71
0.460	2.35	0.510	2.75
0.465	2.39	0.515	2.79
0.470	2.42	0.520	2.84
0.475	2.46	0.525	2.89
0.480	2.50	0.530	2.94
0.485	2.54	0.535	2.99
0.490	2.58	0.540	3.04
0.495	2.62	0.545	3.09
		0.550	3.15

$K_Q = K_{IC}$, if the calculated value of 2.5 $(K_Q/ y.s.)^2$ is less than both the thickness and the crack length of the specimen and provided prior conditions of precracking, temperature, loading rate etc. are met. Whenever the sample fails to meet the requirement, the contractor shall refer the results to the contracting officer.

90.2 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimens is given in ASTM E 399 and may be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT

100.1. The report shall include the following for each specimen tested:

100.1.1. All specimen dimensions measured.

100.1.2. Span between support rollers.

100.1.3. The maximum load P_{max} and the minimum pop-in load P_p, if applicable.

100.1.4. Fracture toughness ($K_Q = K_{1c}$).

100.1.5. Test temperature.

100.1.6. The maximum stress intensity $K_{f(max)}$ during the final 0.020 in. of fatigue cracking.

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100.1.6. Crack plane orientation.

100.1.7. Crack plane orientation.

100.1.8. Fracture appearance.

100.1.8. 0.2 percent offset tensile yield strength.

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